

# PSLE MATHEMATICS PRACTICE PAPER 5

## Algebra & Equations

Duration: 1 hour

Total Marks: 50

Name: \_\_\_\_\_

Date: \_\_\_\_\_

### SECTION A: Short Answer Questions (20 marks)

1. Solve:  $x + 5 = 12$  (2 marks)

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2. Solve:  $2x = 18$  (2 marks)

\_\_\_\_\_

3. Simplify:  $3a + 2b + a - b$  (2 marks)

\_\_\_\_\_

4. Expand:  $3(x + 4)$  (2 marks)

\_\_\_\_\_

5. Solve:  $2x - 3 = 7$  (2 marks)

\_\_\_\_\_

6. Write an expression for 'twice a number plus 5'. (2 marks)

\_\_\_\_\_

7. Solve:  $x/4 = 5$  (2 marks)

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8. Expand and simplify:  $2(x + 3) + 3(x - 1)$  (3 marks)

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## SECTION B: Long Answer Questions (30 marks)

9. Solve:  $3x + 4 = 16$  (4 marks)

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10. A number multiplied by 5 gives 35. Find the number (2 marks)

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11. Simplify:  $5x - 2x + 3y + 2y$  (2 marks)

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12. The perimeter of a rectangle is 24 cm. If length is  $x$  and width is  $(x-2)$ , find  $x$ . (4 marks)

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**13.** Expand:  $(x + 2)(x + 3)$  (4 marks)

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**14.** Solve:  $2(x - 1) = 10$  (3 marks)

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**15.** A boy has  $x$  apples. After giving away 8, he has 12 left. Form and solve an equation. (4 marks)

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## SOLUTIONS

### SECTION A SOLUTIONS

**Q1:  $x + 5 = 12$**

$x = 12 - 5 = 7$

**Q2:  $2x = 18$**

$x = 18 \div 2 = 9$

**Q3: Simplify  $3a + 2b + a - b$**

$3a + a + 2b - b = 4a + b$

**Q4: Expand  $3(x + 4)$**

$3x + 12$

**Q5:  $2x - 3 = 7$**

$2x = 10, x = 5$

**Q6: Twice a number plus 5**

$2x + 5$

**Q7:  $x/4 = 5$**

$x = 5 \times 4 = 20$

**Q8: Expand  $2(x+3) + 3(x-1)$**

$2x + 6 + 3x - 3 = 5x + 3$

### SECTION B SOLUTIONS

**Q9:  $3x + 4 = 16$**

$3x = 12, x = 4$

**Q10: Number  $\times 5 = 35$**

Number  $= 35 \div 5 = 7$

**Q11: Simplify  $5x - 2x + 3y + 2y$**

$3x + 5y$

**Q12: Rectangle perimeter**

$2(x + x - 2) = 24, 2(2x - 2) = 24, 2x - 2 = 12, x = 7$

**Q13: Expand (x+2)(x+3)**

$$x^2 + 3x + 2x + 6 = x^2 + 5x + 6$$

**Q14:  $2(x-1) = 10$**

$$2x - 2 = 10, 2x = 12, x = 6$$

**Q15: Apples equation**

$$x - 8 = 12, x = 20$$

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